

A Comparative Study of AI Models In Software Development

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1 Introduction

In the last few years artificial intelligence(AI) has become more present within software development as its capabilities has improved. An often discussed issue with AI presence in software development is the reliability of the solutions it suggests. For this reason, the projects goal is to answer 3 key questions related to the before mentioned issue. 1.) Which model provides superior solutions, 2.) How reliable are the solutions when the complexity of the problem we are trying to solve increases and 3.) How well can AI be trusted overall when it comes to software development.

2 Related work

A paper by Russel Poldrack, Thomas Lu and Gasper Begus which covers AI-assisted coding with the GPT-4 model, that while it was a powerful tool, it had shortcomings in terms of validity and suggested that humans where still required to ensure the validity of the AI generated code. Considering the paper is written in 2023, this creates the question of whether this has changed as AI models have become more advanced or if their findings from 2023 still holds true. While searching for literature relevant to the project, very few papers matched the purpose of the project, the referenced paper being the only one found which was directly relevant to the goals of the project. A majority of the papers found focused mainly on AI as a whole and not specifically as coding assistants. It is therefore difficult to say how heavily researched this topic is, as a wider and more extensive search could yield more relevant results or it could be a dead end. Either way, the lack of more relevant results while searching for papers could indicate that the specific field of the project is not as widely researched as other aspects concerning AI, which in turn could make the project a usefull contribution.

3 System

The planned system to study are how well different AI models manages to provide good solutions in writing increasingly complex algorithms. Therefore, the system is limited to the code provided by the model. The idea for determining whether a solution is ideal or not, will be the overall correctness, if redundant code is present, and execution time. These three metrics will then be evaluated in such a way that a minor disadvantage in say execution time does not disqualify a model if the other two metrics yield good results. For benchmarking the execution time, simple utilities will be implemented. Due to their popularity, two of the models for the experiment will be Claude and GPT. The idea is to also have a third model be part of the experiments, however, exactly which one is not yet decided. The choice will depend on popularity within coding (not necessarily overall popularity) as this will be more relevant to the purpose of the project. To ensure fairness, all models will be given the exact same prompt, while this might inadvertently benefit models which are overall more heavily used within software development and as such might have a larger database, it is the only obvious method of ensuring that the experiments are as fair as possible. This also allows for the possibility of measuring the level of "handholding" required by the user.

4 Early environment and scope

Due to the projects nature, requirements in terms of hardware and software is fairly limited. Different hardware might be used for different experiments, however, all comparative benchmarks will run on the same hardware as to not give any benefits in terms of running on better hardware. Exactly which algorithms will be used for the experiments is not yet decided, meaning dataset requirements is currently unknown. For the experiments one of the two following sets of hardware will be used:

Component	Rig 1	Rig 2
CPU	Intel Core I5 9600k	Intel Core I5 10300H
RAM	16GB DDR4	8GB DDR4
GPU	AMD Radeon 6700XT 12 GB	Nvidia Geforce GTX 1650 4 GB

Like previously mentioned, all comparative benchmarks and other tests will be ran on the same hardware, which of the ones will be used for a given experiment comes down to convenience at the time. The GPU is listed for the sole reason of giving a more clear view of the accessible hardware for the experiments, most if not all algorithms used for experiments will be purely CPU intensive. It is also worth mentioning that with rig nr 1s current configuration in WSL, the dedicated GPU is not accessed and instead the CPUs onboard graphics is used. The hardware configuration used during an experiment will also be mentioned in the relevant report.

5 Early timeline

Initial work will consist of first deciding which AI models are going to be part of the experiment, and deciding exactly which algorithms should be used. As previously mentioned, execution time will be a relevant metric during experiments, therefore, a benchmarking framework will be implemented relatively early in the project. The exact design of the framework is not yet decided, as this might change depending on the needs of each experiment.

Initial plans for the experiments will revolve around sorting algorithms. The reason for starting with these algorithms are their overall simplicity, as such all of the models are expected to be able to handle this task. Further experiments will gradually increase the complexity of the algorithms, exactly which will be used is currently undecided. However, binary search trees and indexing algorithms are good candidates and are being considered. The plan is to have the sorting experiment done by the deadline for the second assignment which will be the first phase of the project.

6 Discussion

When considering risks of choices, one in particular could prove problematic, token usage. Several models make use of a token system for requests. For non-paid solutions, the risk of running out of tokens when experimenting is a prevalent one. There are mitigating factors, swapping host sites/programs being the solution which will likely solve the issue altogether. Another issue with the project is the fact that with LLMs, the same input can produce different outputs. Addressing this issue completely to the point where every single experiment is going to be 100 percent reproducible might not be possible (at least not without knowing the inner workings of each model). To ensure that the experiments are as reproducible as possible, all LLM interactions such as the prompts used and provided responses will be documented as well as possible.

7 References

Russel A.Poldrack, Thomas Lu, Gasper Begus (2023). AI-assisted coding: Experiments with GPT-4